# InfraRisk AI: Infrastructure Finance Risk Modelling & Credit Assessment Platform

## Technical Report

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### Project

InfraRisk AI – Infrastructure Finance Risk Modelling & Credit Assessment

### Duration

15-Day Data Science Project

### Technologies

Python, Pandas, NumPy, XGBoost, SHAP, Prophet, Streamlit, Pytest, Docker, Scikit-Learn

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# Abstract

Infrastructure projects represent some of the largest and most complex investments globally, requiring rigorous assessment of financial, operational, construction, and macroeconomic risks. Traditional project finance assessments rely heavily on spreadsheet-based analysis and static assumptions, limiting their ability to capture dynamic risk factors and uncertainty.

InfraRisk AI was developed as an end-to-end infrastructure finance risk intelligence platform designed to evaluate project-level risk, forecast future performance, simulate stress scenarios, and provide explainable machine learning insights. The platform integrates credit risk modelling, Monte Carlo stress testing, revenue forecasting, explainable AI, and infrastructure simulation within a unified dashboard environment.

The solution enables financial institutions, infrastructure investors, and project finance analysts to assess probability of default, expected loss, and infrastructure resilience under various economic and operational conditions.

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# 1. Introduction

Infrastructure financing plays a critical role in supporting economic development through transportation networks, airports, ports, renewable energy facilities, telecommunications infrastructure, and public utilities.

Infrastructure projects are exposed to numerous risk factors including:

- Construction delays
- Cost overruns
- Sovereign risk
- Inflation
- Interest rate fluctuations
- Regulatory changes
- Demand uncertainty

The objective of this project was to design and implement an AI-powered platform capable of assessing and forecasting infrastructure project risk through quantitative and explainable methods.

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# 2. Problem Statement

Traditional project finance analysis often suffers from:

- Static assumptions
- Limited scenario testing
- Lack of explainability
- Manual risk assessment processes
- Limited forecasting capabilities

The challenge was to create an integrated platform capable of:

1. Evaluating infrastructure project risk.
2. Predicting default probability.
3. Forecasting project revenues.
4. Performing stress testing.
5. Explaining model predictions.
6. Simulating real-world infrastructure risk events.

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# 3. Objectives

The primary objectives of InfraRisk AI were:

- Build a synthetic infrastructure finance dataset.
- Engineer project-finance specific risk features.
- Develop a machine learning credit risk model.
- Implement SHAP explainability.
- Create Monte Carlo stress-testing capabilities.
- Forecast infrastructure revenues.
- Design an infrastructure risk simulation environment.
- Deliver insights through an interactive dashboard.

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# 4. Dataset Design

A synthetic infrastructure finance dataset containing 10,000 infrastructure projects was generated.

Project categories included:

- Transportation
- Airports
- Ports
- Renewable Energy
- Telecommunications

Dataset variables included:

- Project Cost
- Debt Ratio
- Interest Rate
- Construction Delay
- Cost Overrun Percentage
- GDP Growth
- Inflation Rate
- Sovereign Risk Score
- Traffic Demand Index
- DSCR
- LLCR
- PLCR

The dataset was specifically designed to reflect project-finance concepts discussed in infrastructure financing literature.

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# 5. Feature Engineering

Feature engineering was performed to transform raw project data into meaningful risk indicators.

Key engineered features included:

### Leverage Risk

Measures project exposure to debt financing.

### Construction Risk Score

Combines:

- Schedule delays
- Cost overruns

### Macroeconomic Risk Score

Combines:

- Inflation
- GDP Growth

### Financial Strength

Calculated using:

- DSCR
- LLCR
- PLCR

### Credit Risk Metrics

- Probability of Default (PD)
- Loss Given Default (LGD)
- Exposure at Default (EAD)
- Expected Loss (EL)

### Composite Risk Score

A weighted infrastructure risk indicator combining:

- Construction risk
- Macroeconomic risk
- Sovereign risk
- Financial strength

# 6. Credit Risk Modelling

## Model Selection

XGBoost was selected due to its strong performance on structured tabular datasets and ability to model non-linear relationships.

Advantages:

- High predictive accuracy
- Robust handling of feature interactions
- Built-in feature importance
- Scalability

## Target Variable

The target variable represented infrastructure project default risk using project finance indicators.

## Model Evaluation

The model achieved:

- Accuracy: 98.25%
- ROC-AUC: 0.989

Additional evaluation metrics included:

- Confusion Matrix
- Classification Report
- Precision
- Recall
- F1 Score

Results demonstrated strong predictive capability for infrastructure project risk classification.

# 7. Explainable AI using SHAP

Machine learning explainability is critical within financial decision-making environments.

SHAP (SHapley Additive Explanations) was implemented to provide transparency into model predictions.

Benefits:

- Feature contribution analysis
- Credit committee transparency
- Regulatory compliance support
- Risk driver identification

Outputs included:

- SHAP Summary Plot
- Global Feature Importance Analysis

The SHAP analysis highlighted major contributors to project default risk including:

- Construction delays
- Cost overruns
- Debt ratio
- Sovereign risk
- Macroeconomic variables

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# 8. Monte Carlo Stress Testing

A Monte Carlo simulation framework was developed to evaluate project resilience under uncertainty.

## Simulated Variables

- GDP Growth Shock
- Inflation Shock
- Interest Rate Shock
- Traffic Demand Shock
- Cost Overrun Shock

## Simulation Volume

10,000 scenarios

## Outputs

- Probability of Default
- Expected Loss
- DSCR Distribution
- Stress Scenario Results

Results:

- • Probability of Default: 7.30%
- • Expected Loss: $6.39 Million
- • Average DSCR: 1.23

The simulation engine enabled dynamic stress testing beyond traditional deterministic financial modelling.

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# 9. Revenue Forecasting

Revenue forecasting was implemented using Meta Prophet.

## Forecasting Process

Historical infrastructure revenue series were generated and used to train the forecasting model.

Outputs included:

- • Future Revenue Projections
- • Confidence Intervals
- • Trend Analysis

The forecasting module supports long-term infrastructure investment planning and financial analysis.

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# 10. InfraRisk Lab Simulation

A gamified infrastructure finance simulation environment was developed.

Users can select:

- • Toll Road
- • Airport
- • Solar Plant
- • Port
- • Telecom Tower

Random infrastructure events are generated including:

- • Construction Delays
- • Inflation Shocks
- • Sovereign Downgrades
- • Regulatory Changes
- • Demand Collapses

The simulation engine recalculates project risk and provides mitigation recommendations.

This component transforms static risk analysis into an interactive decision-support experience.

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# 11. Dashboard Development

A Streamlit dashboard was developed to integrate all project components.

Dashboard modules include:

### Executive Overview

Portfolio-level metrics and KPIs.

### Credit Risk Analytics

Project risk analysis and category distribution.

### SHAP Explainability

Feature contribution visualizations.

### Monte Carlo Analysis

Stress-testing outputs and DSCR distributions.

### Revenue Forecasting

Future revenue projections and uncertainty intervals.

### InfraRisk Lab

Interactive infrastructure simulation environment.

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# 12. Testing and Validation

Pytest was used for software validation.

Test coverage included:

- Feature engineering validation
- Expected loss calculation

- Risk score validation
- Monte Carlo engine validation
- Simulation engine validation

Result:

5 automated tests passed successfully.

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## 13. Docker Deployment

Docker was used to containerize the application and ensure deployment portability.

Benefits:

- Environment consistency
- Simplified deployment
- Reproducibility
- Scalability

The dashboard was successfully deployed and executed within a Docker container.

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## 14. Key Results

| Metric | Result |
|---|---|
| Projects Analysed | 10,000 |
| Features Engineered | 31 |
| Model Accuracy | 98.25% |
| ROC-AUC | 0.989 |
| Monte Carlo Simulations | 10,000 |
| Probability of Default | 7.30% |
| Expected Loss | $6.39M |
| Tests Passed | 5 |
| Dashboard Modules | 6 |

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# 15. Limitations

The project utilized synthetic data and therefore may not fully represent real-world infrastructure finance dynamics.

Additional limitations include:

- Limited historical data availability
- Simplified sovereign risk modelling
- Simplified revenue generation assumptions
- No real-time data integration

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# 16. Future Enhancements

Future improvements may include:

- Graph Neural Networks (GNNs)
- Temporal Fusion Transformers (TFTs)
- Satellite Imagery Analysis
- Reinforcement Learning Debt Optimization
- NLP Contract Intelligence
- Portfolio Optimization
- Real-Time Infrastructure Monitoring

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# 17. Conclusion

InfraRisk AI successfully demonstrates how machine learning, explainable AI, forecasting, simulation, and quantitative risk analytics can be integrated into a unified infrastructure finance intelligence platform.

The solution provides a practical framework for evaluating infrastructure project risk, forecasting future performance, stress testing under uncertainty, and supporting financial decision-making. Through the integration of XGBoost, SHAP, Monte Carlo simulation, Prophet forecasting, and interactive simulation capabilities, InfraRisk AI offers a scalable and extensible foundation for next-generation infrastructure risk assessment systems.

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# References


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2. World Bank Infrastructure Finance Reports.
3. IMF Economic Outlook Database.

4. SHAP Documentation.
5. XGBoost Documentation.
6. Prophet Documentation.
7. Project Finance Principles and Practices.
8. Infrastructure Investment and Risk Management Literature.